

Cryptography - 21_11

Last construction that I can get, this time from DL assumption, is a CRH.
Here is a simple construction over $(\mathbb{G}, g, q)$, with q a prime.

$$H_{g_1, g_2}(x_1, x_2) = g_1^{x_1} g_2^{x_2} \in \mathbb{G}$$

Example

$\mathbb{G} = \mathbb{QR}_p$ such that $\frac{p-1}{2} = q$

where $g_1 = g$ (the generator) and g_2 any group element

Given a collision (x_1, x_2) and (x'_1, x'_2) such that $(x_1, x_2) \neq (x'_1, x'_2)$ and

$$g_1^{x_1} g_2^{x_2} = g_1^{x'_1} g_2^{x'_2}$$

$$g_1^{x_1 - x'_1} = g_2^{x_2 - x'_2}$$

$$g_2 = g_1^{(x_1 - x'_1)(x_2 - x'_2)^{-1}}$$

We have $x_2 \neq x'_2$, then the inverse exists.

(Note that if $x_2 = x'_2$ then also $x_1 = x'_1$)

$$\implies (x_1 - x'_1)(x_2 - x'_2)^{-1} \text{ is the DL of } g_2$$

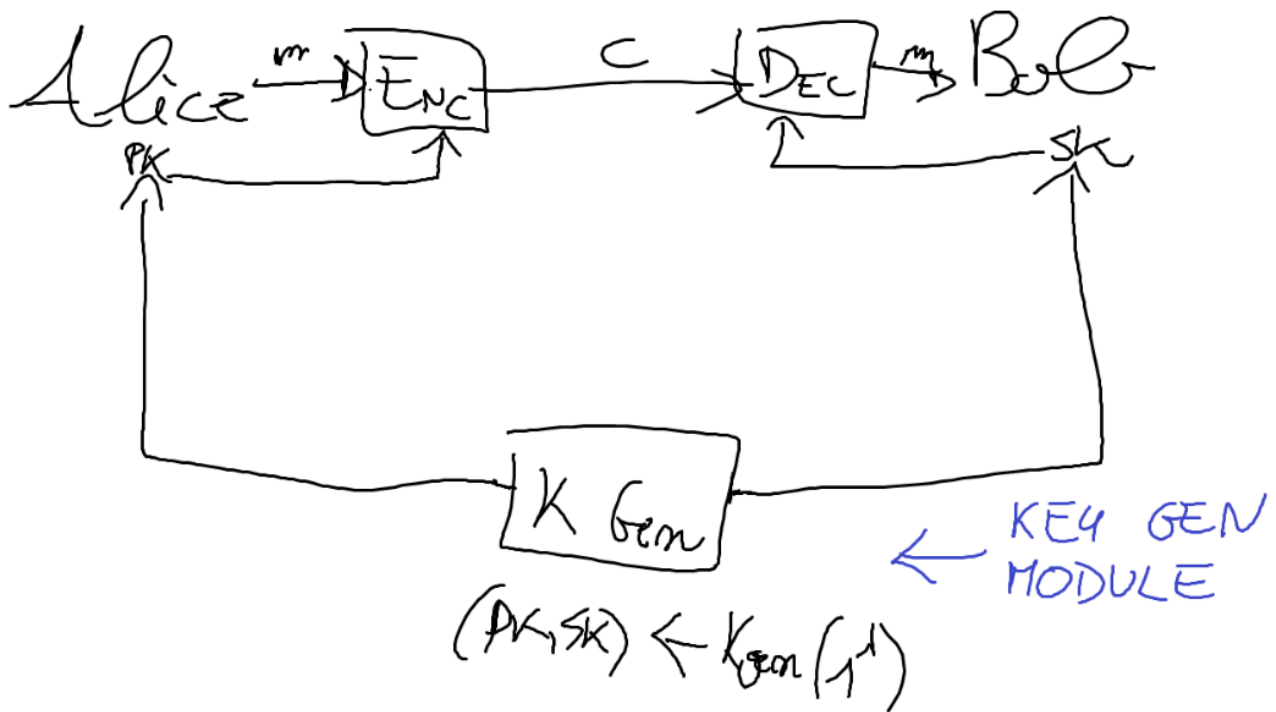
Let's build the reduction:



As $g_2 = y$ and $(x_1 - x'_1)(x'_2 - x_2)^{-1}$ is the x for the DL game for $g_2 = y$.

Public-key Encryption

This came after the DDH protocol.



The key gen module outputs a secret key for Bob, and a public key for Alice.

We are neglecting a problem: Alice should know pk is the public key of Bob. How to do that? Digital signatures and public key infrastructure.

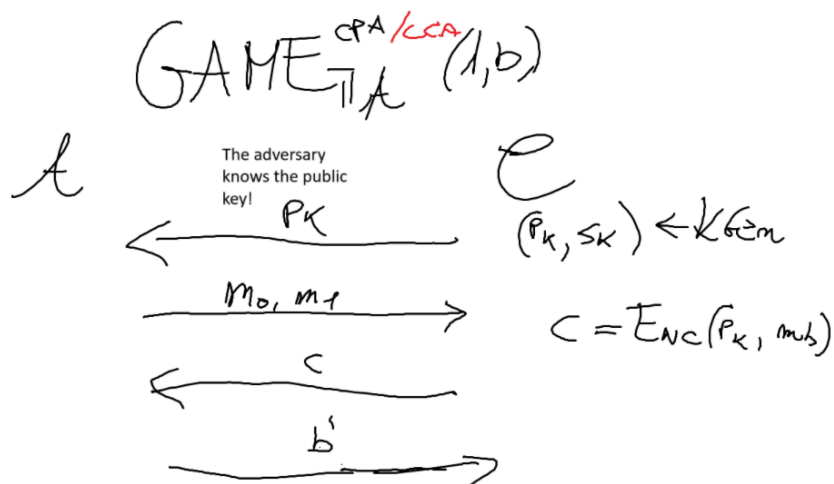
Security: CPA/CCA.

✓ Definition

$\Pi = (KGen, Enc, Dec)$ is CPA/CCA secure if

$$GAME_{\Pi, A}^{cpa/cca}(\lambda, 0) \approx_c GAME_{\Pi, A}^{cpa/cca}(\lambda, 1).$$

What is the game?

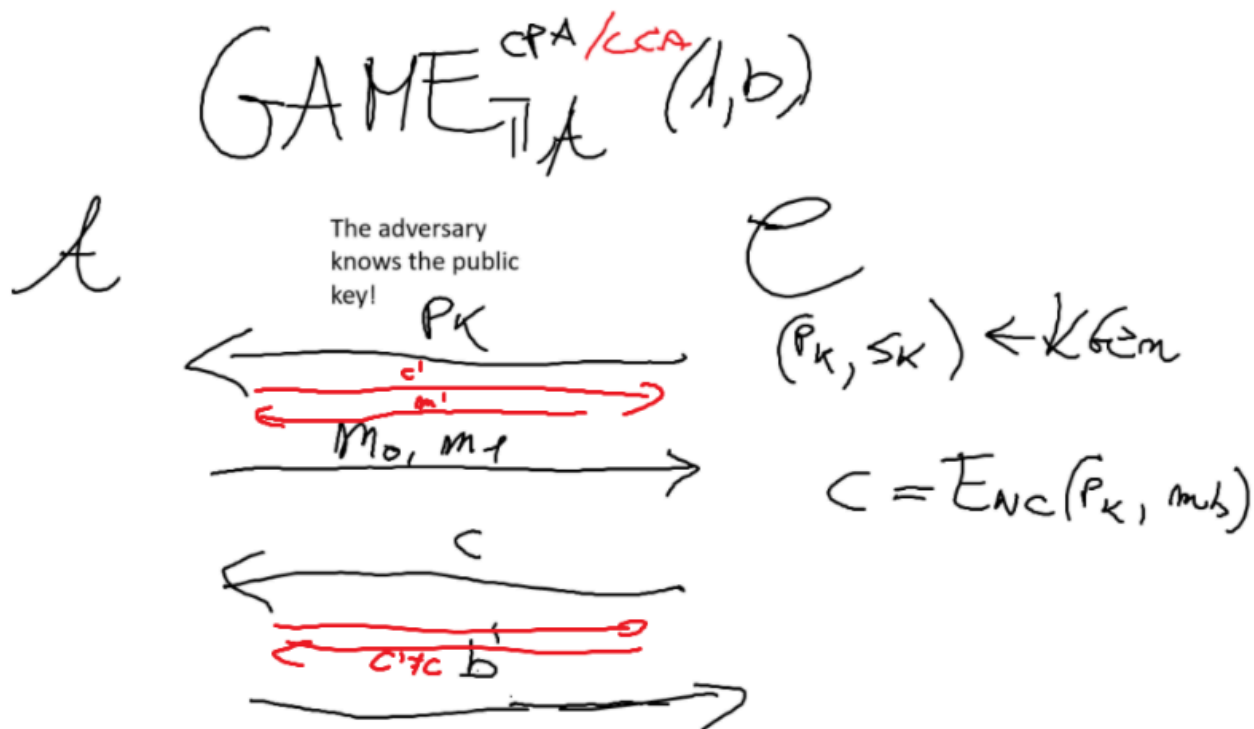


The adversary is now able to encrypt whatever he wants, as he has access to the public key.

Consider that the attacker is basically Bob in the scheme above, while the challenger is Alice, so $\mathcal{C}$ encrypts with the public key.

Obviously this works if the encryption scheme is not deterministic, as in that case $\mathcal{A}$ could simply compute itself c_0 and c_1 and compare them with the received c .

For CCA, we have this modification:



We'll see two constructions:
ElGamal and RSA.

Let's start with **ElGamal**.

We'll have:

- Public parameters: $(\mathbb{G}, g, q)$
- Key gen: $pk = h = g^x$, $sk = x$, $x \leftarrow \mathbb{Z}_q$
- Encryption: Pick $r \leftarrow \mathbb{Z}_q$, and then we output $c = (c_1, c_2) = (g^r, h^r \cdot m)$, with $m \in \mathbb{G}$
- Decryption: $\frac{c_2}{c_1^x} = \frac{h^r \cdot m}{g^{r \cdot x}} = \frac{\cancel{h^r} \cdot m}{\cancel{h^r}} = m$

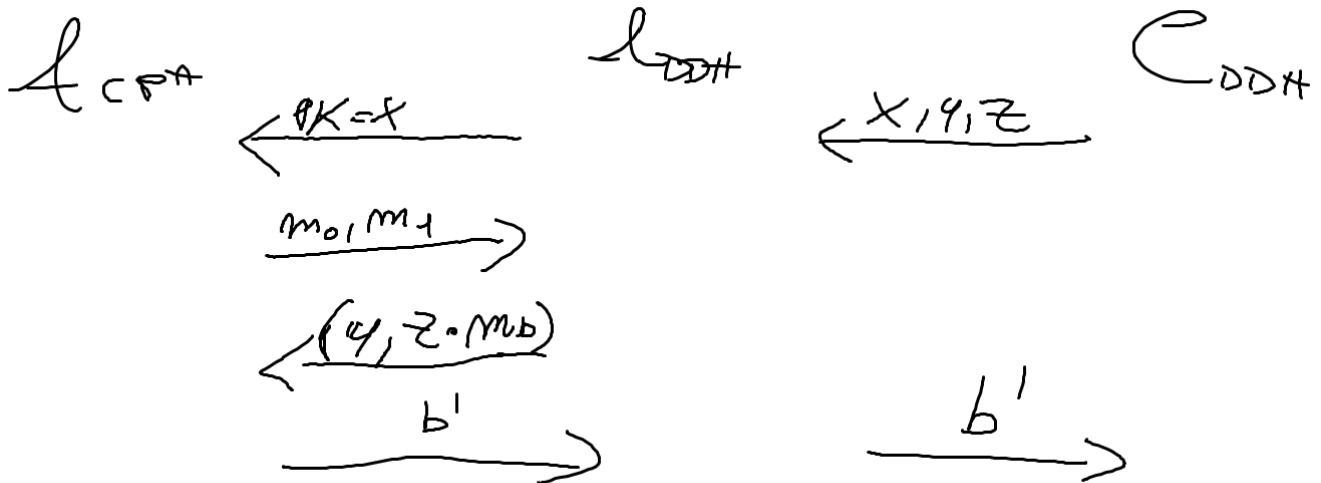
✓ **Theorem**

The above PKE is CPA-secure under DDH

Proof

Start with $G(\lambda, b)$ the CPA game and define $H(\lambda, b)$ where h^r is replaced by g^z for $z \leftarrow \mathbb{Z}_q$.

Now: $G(\lambda, b) \approx_c H(\lambda, b) \forall b$ by DDH assumption.



Either:

$$(X, Y, Z) = \begin{cases} g^x, g^r, g^{x \cdot r} \\ g^x, g^r, g^z \end{cases}$$

In the first case the reduction simulated $G(\lambda, b)$, in the second $H(\lambda, b)$.

On the other hand: $H(\lambda, 0) \equiv H(\lambda, 1)$.

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